

WEBSITE USER GUIDE

WEB-BASED GEOGRAPHIC INFORMATION SYSTEM

FOR FLOOD RISK MAPPING AREA ANALYSIS

This document is compiled as a user guide for the web-based Geographic Information System (GIS) website for flood risk analysis developed using Remote Sensing and WebGIS technology. This guide is intended for general users and disaster management staff so that they can use the system optimally.

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CHAPTER I

INTRODUCTION

1.1 Background

Floods are one of the natural disasters that have a significant impact on people's lives. Therefore, a system that can provide spatial information quickly, accurately, and easily is needed. This GIS website was developed to visualize and analyze flood risks using remote sensing data.

1.2 Purpose of the Guide

This guide aims to provide technical instructions on how to use the flood risk analysis GIS website so that users can understand the functions, features, and usage flow of the system properly.

1.3 Scope

The scope of this guide includes how to access the website, an explanation of the user interface, and steps for using the main features.

General Public:

- Needs: Requires flood risk information that is easy to understand without technical expertise and without the need to install heavy software.
- Solution Scope: The system is built on a web-based platform (WebGIS) so that it can be accessed directly through a browser on a smartphone or laptop. The interface is designed to be user-friendly so that the public can interactively view maps of vulnerable areas and the locations of public facilities.

Government Agencies/Disaster Mitigation:

- Requirements: Requires accurate data to support evacuation decision-making or infrastructure planning.
- Solution Scope: Provides an analysis dashboard that integrates monthly/daily rainfall data, topography, and historical data for precise trend analysis.

CHAPTER II

OVERVIEW OF THE SYSTEM

2.1 System Description

This web-based Geographic Information System was developed with a client-server architecture. The backend uses Flask as a proxy server connected to Google Earth Engine, while the frontend uses ReactJS and Leaflet for map visualization.

2.2 User Access Rights

This website can be accessed by the general public without authentication. Users can view maps, set layers, and perform spatial analysis.

CHAPTER III

WEBSITE USER GUIDE

3.1 Website Access

Users can access the SIG website through web browsers such as Google Chrome or Mozilla Firefox by typing in the website's domain link. Make sure you have a stable internet connection.

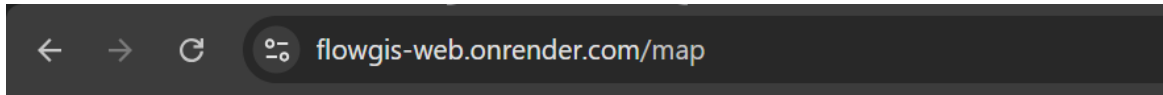


Figure of Accessing Websites Through a Browser

3.2 Manual Book

When successfully accessing the website link, users will find a notification to read the manual. Users can choose to read the manual or skip it.

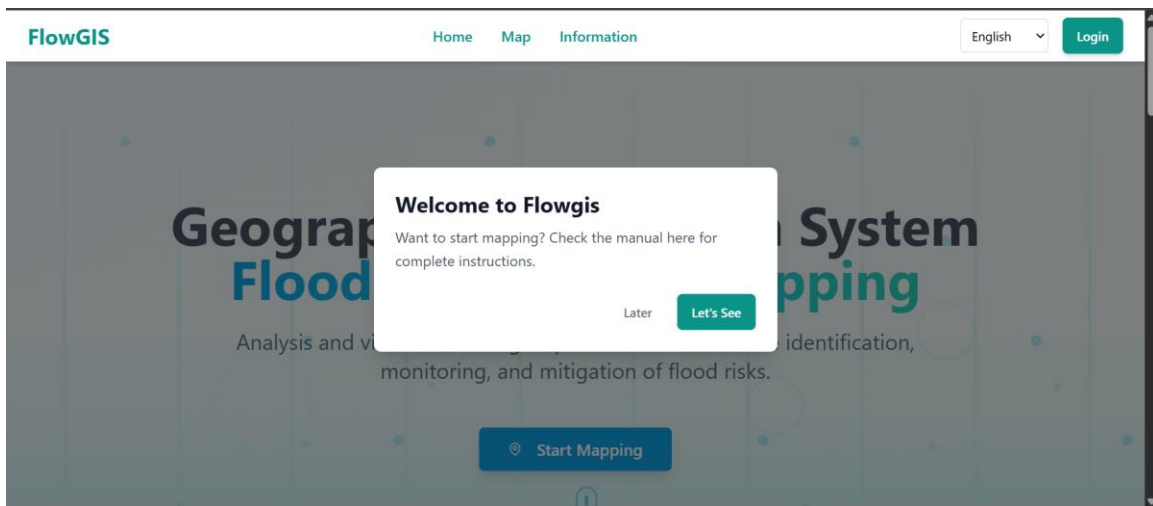


Figure of Manual Book Reading Notification

3.3 Landing Page

The landing page is the initial display that presents brief information about the website, containing several related features, contact information, and a message form. This page also includes navigation to the map and information pages.

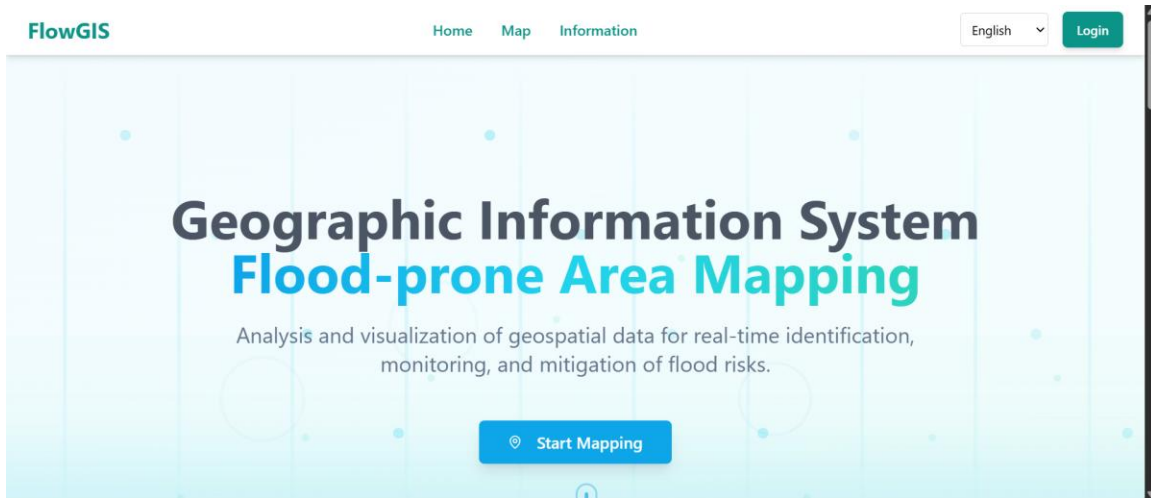


Figure of Landing Page

3.4 Change Language

Users can change the language on the website by simply clicking on the dropdown menu, selecting Language, and then clicking on the desired language. This website supports 3 languages.

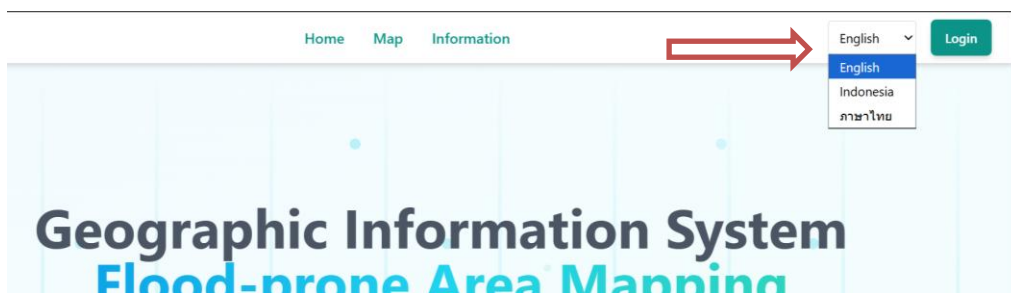


Figure of Change Language Feature

3.5 Login and Register (Sign in)

Users can access the map without logging in or in anonymous mode. However, to access some features, users need to log in. Users only need to log in by pressing the login

button and filling in the required data correctly. If users do not have an account yet, they can register first and then log in again. If the login is successful, a notification will appear and users will be directed to the web page.

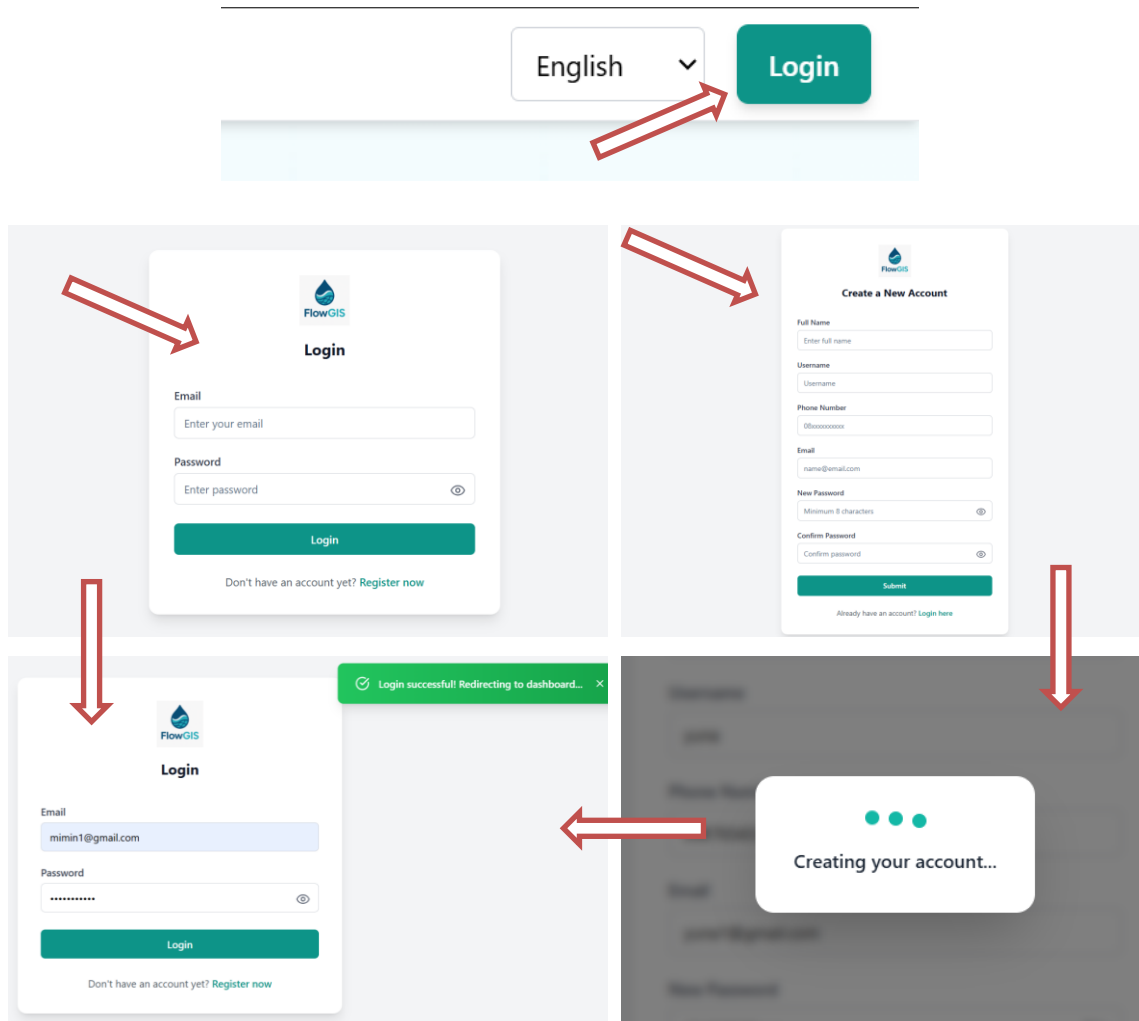


Figure of Login and Registration Steps

3.5 Map Page (Feature introduction)

After successfully logging in or entering anonymous mode, users can access the map page. The map page contains several components and features, which will be explained in the following image.

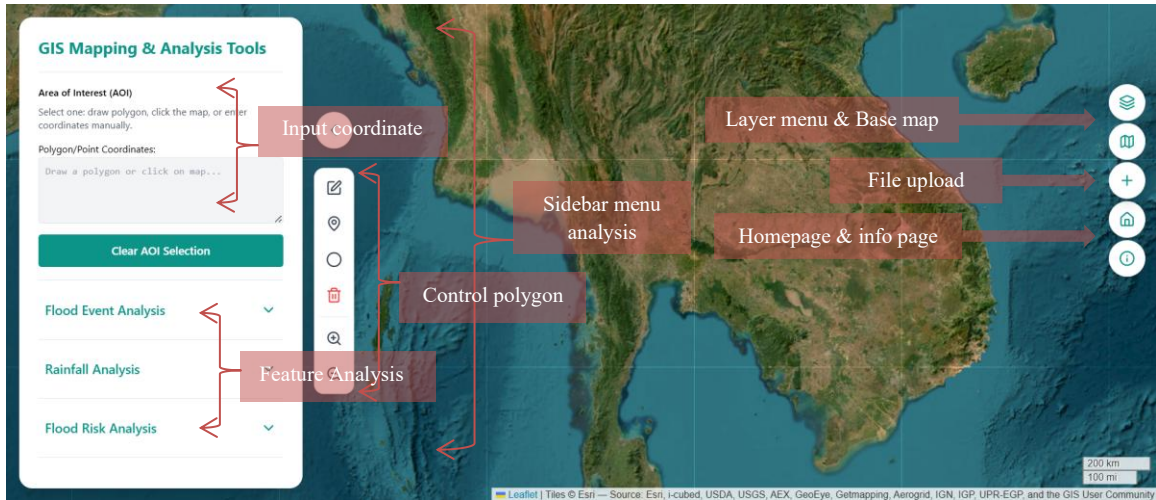
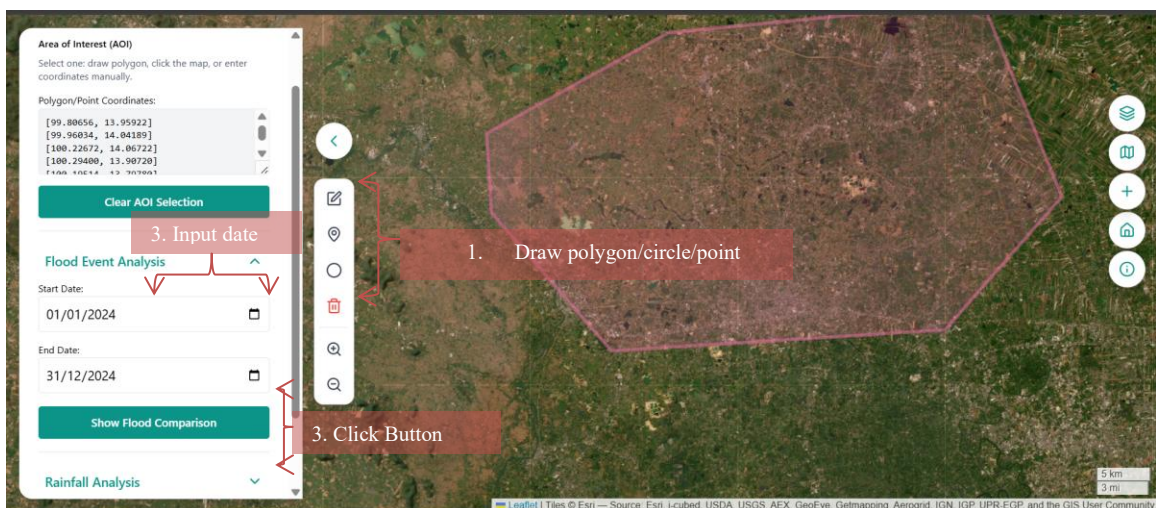


Figure of Map Page

3.6 Flood Event Analysis

Users can analyze flood events within a desired time period. Users can draw polygons by clicking on draw polygon, draw circle, or point. After that, enter the start date and end date. Then click the button to see the flood comparison.



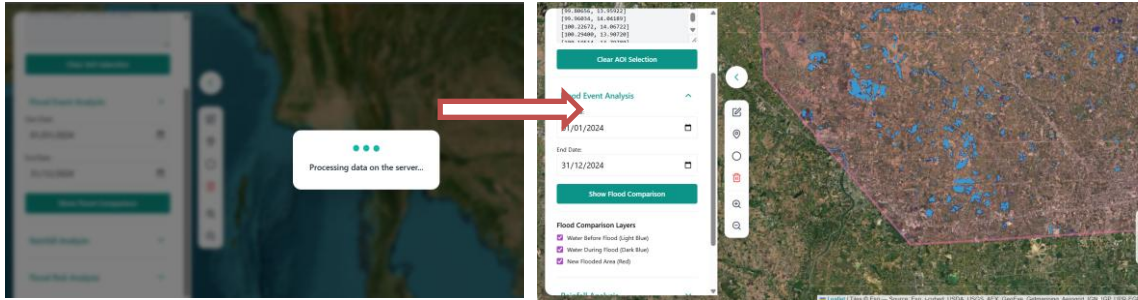


Figure of Steps for Using the Flood Event Analysis Feature

3.6 Rainfall Analysis

For rainfall analysis, follow the same steps as in the Flood Event Analysis feature. Start by drawing or creating a polygon, select the analysis mode (Monthly/Daily), specify the start and end times, and click the button to display the graph and maps.

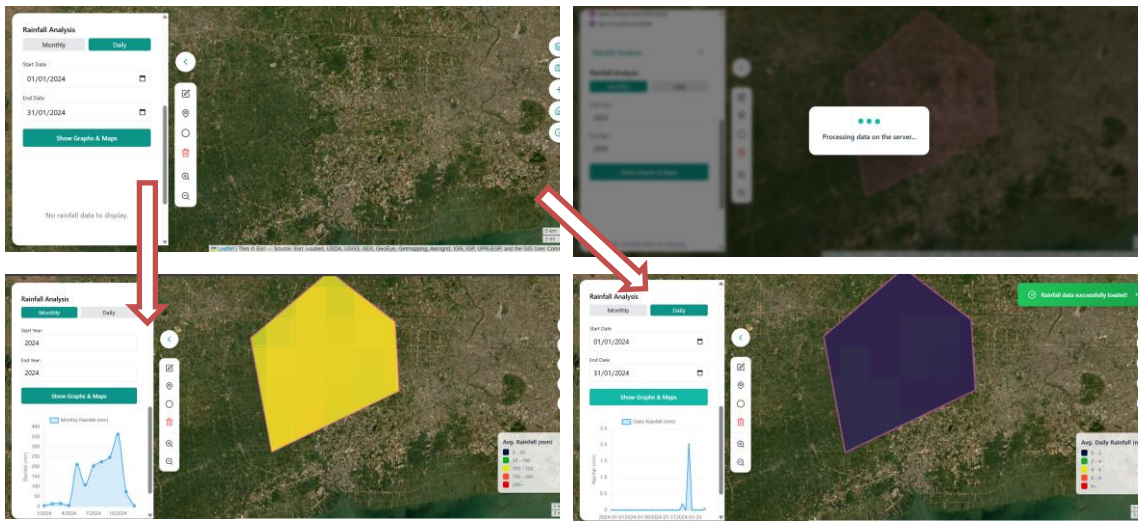


Figure of Steps for Using the Rainfall Analysis Feature

3.7 Flood Risk Analysis

For risk analysis analysis, follow the same steps as in the Flood Event Analysis feature. Start by drawing or creating a polygon, input the start and end years of the analysis, and click the analysis button. To view the flood susceptibility layer, users can click the checkbox to display it on the map.

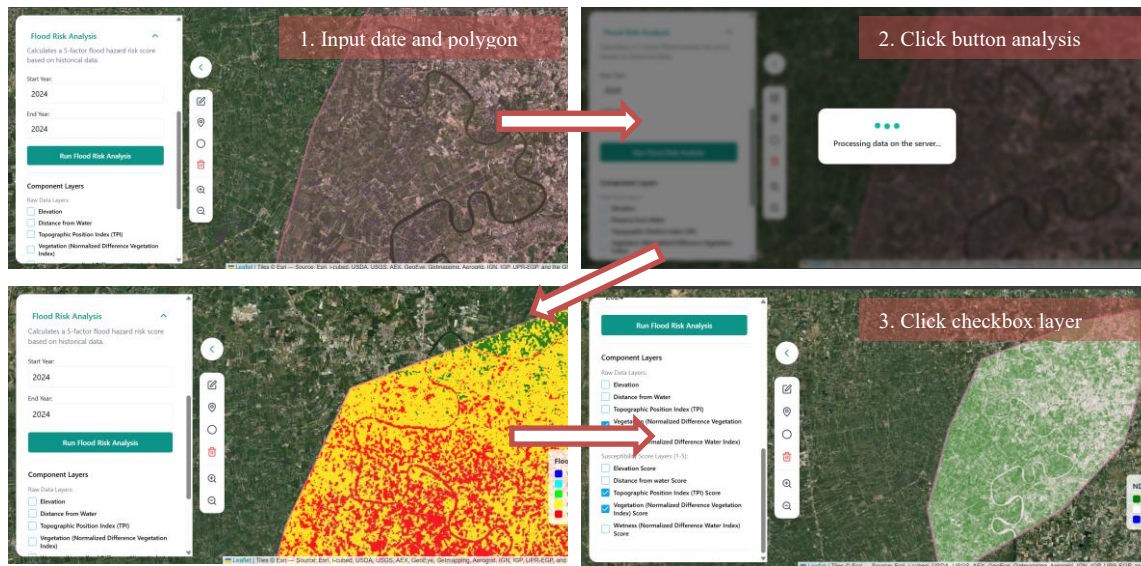


Figure of Steps for Using the Flood Risk Analysis Feature

3.8 Map Layer Control

Users can enable or disable map layers such as administrative boundaries, public facilities, rainfall, and flood-prone areas through the layer control panel. Users click on the layer icon and select the type of layer they want to view.

This feature also provides various layers that support geosocial mapping in the Nakhon Pathom area. This information can be used as a basis for decision-making in mitigation and water management.



Figure of Steps for Using the Layer Control Feature

3.9 Upload File Area of Interest

Users who do not want to draw polygons manually can upload files in the following formats: GeoJSON, KML, Shapefile (ZIP). Users need to click the upload button and upload files in the appropriate format. After successfully uploading, users can perform analysis.

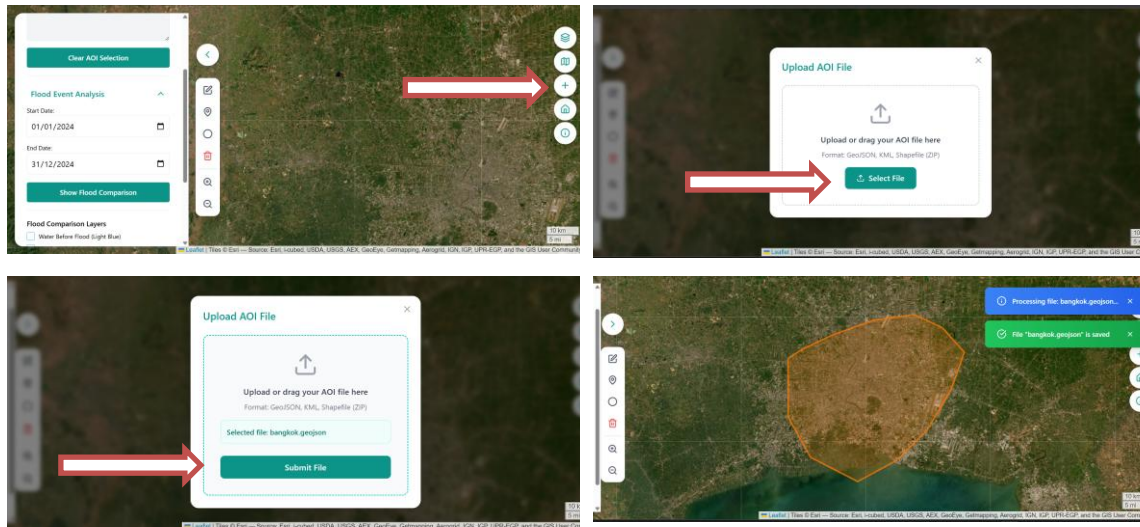


Figure of Steps for Using the Layer Control Feature

CHAPTER IV

CONCLUSION

4.1 Conclusion

This guidebook is expected to assist users in understanding and operating the flood risk analysis GIS website effectively. Further development is recommended to include the integration of mobile platforms and Internet of Things (IoT)-based monitoring systems.